

ANTHROPOGENIC THREATS TO INTERMITTENT RIVERS AND EPHEMERAL STREAMS

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Ming-Chih Chiu^{*,†}, Catherine Leigh^{‡,§,¶}, Raphael Mazor^{**}, Núria Cid^{††}, Vincent Resh^{*}

University of California, Berkeley, CA, United States^{} National Chung Hsing University, Taichung City, Taiwan[†]
Irstea, UR MALY, centre de Lyon-Villeurbanne, Villeurbanne, France[‡] CESAB-FRB, Immeuble Henri Poincaré,
Aix-en-Provence, France[§] Griffith University, Nathan, QLD, Australia[¶] Southern California Coastal Water Research
Project, Costa Mesa, CA, United States^{**} Universitat de Barcelona (UB), Barcelona, Spain^{††}*

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- In intermittent rivers and ephemeral streams (IRES), significant anthropogenic threats result from alterations in natural flow regimes, changes in geomorphological features, and degraded water quality. These changes may favor invasion by alien species, a major ecological problem in IRES.
- Anthropogenic threats in IRES often exceed the thresholds of native species' adaptations and can change community structure and ecological processes. Impacts may cascade to downstream waterbodies and adjacent riparian wetlands.
- Hydrology, geomorphology, and water quality impacts often occur concurrently (e.g., from mining activities) and result in multiple stressors in IRES, which can increase their vulnerability to alien species.
- IRES are under-appreciated in terms of the ecosystem services they provide, resulting in a lack of biological monitoring systems which consequently hinders effective management strategies.

5.1.1 INTRODUCTION: THE MAIN THREATS TO IRES

Intermittent rivers and ephemeral streams (IRES) occur worldwide and make important contributions to freshwater biodiversity and biogeochemical cycles at local, catchment, regional, and global scales (Chapters in Sections 1–4; [Datry et al., 2014](#)). IRES in all parts of the world are subject to many of the same human-induced threats as perennial rivers and streams. However, most IRES have far less legal protection than their perennial counterparts ([Nikolaidis et al., 2013](#); [Acuña et al., 2014](#); [Chapter 5.3](#)), reflecting the low value that society places on their ecological attributes and ecosystem services ([Chapter 5.2](#)). Given this lack of appreciation, IRES are probably under greater threat of degradation than perennial systems. They can serve as sites for the dumping of trash and dredging of sediment or as conduits for waste water disposal and road traffic ([Fig. 5.1.1](#)). Furthermore, the natural variability of their flow regimes and hydrological phases, which can result in extended and unpredictable periods of zero flow ([Chapters 2.2 and 2.3](#)), means that these activities, along with anthropogenic changes

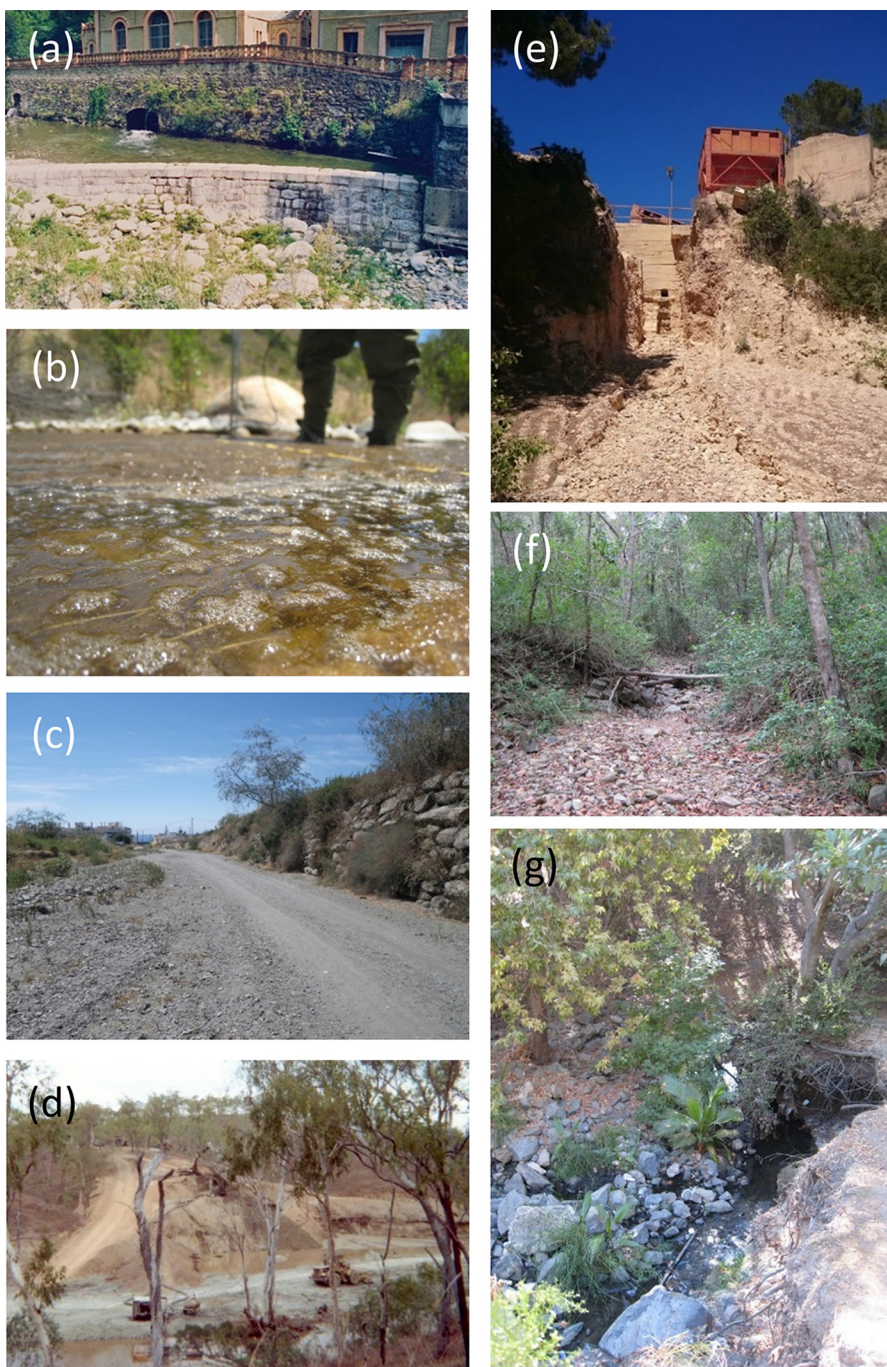


FIG. 5.1.1

See legend on opposite page.

to their hydrology induced by artificial dewatering or augmented flows (Fig. 5.1.1), may be wrongly dismissed as unproblematic.

Flow intermittence and the shifting aquatic-terrestrial habitat dynamics of IRES (Chapter 4.9) present challenges to both the aquatic and terrestrial biota inhabiting these systems. The biota must either withstand the episodic unfavorable conditions such as dry habitat for aquatic biota or flooded habitat for terrestrial biota in situ (i.e., exhibit resistance) or escape them to return later when conditions again become favorable (i.e., exhibit resilience). Many taxa that inhabit IRES thus possess adaptive traits of resistance and/or resilience to flow extremes, including drying and flooding (e.g., Bêche et al., 2006; Leigh et al., 2016a; Chapter 4.8). However, anthropogenic activities that alter the natural flow regimes, geomorphology, and water quality of IRES, and even the composition of biotic communities (e.g., via introductions of alien species or clearing of vegetation) may push these ecosystems and biota beyond their adaptation thresholds.

In this chapter, we review anthropogenic threats to IRES and their biota, broadly classified into the following (Fig. 5.1.2): hydrological alterations, including withdrawals or additions of water that change inundation durations and the dynamic interplay of aquatic vs terrestrial habitat (as well as altering other ecologically relevant aspects of the flow regime (Poff et al., 2010)); physical and chemical alterations, including sedimentation, sediment mining, and water pollution that degrade aquatic and terrestrial habitats in IRES; and biological alterations, in particular the introduction or invasion of species that can threaten native communities. We discuss these threats and their individual and interactive effects on IRES, both those that undergo natural flow intermittence and those once-perennial rivers that now undergo anthropogenically induced intermittence and their adjacent ecosystems. Furthermore, we consider how climate change may interact with these threats or even induce hydrological, physical, chemical, and biological alterations directly (Box 5.1.1). Finally, we consider future needs and challenges for research and management to effectively prevent, reduce, or mitigate the impacts of anthropogenic alterations to IRES ecosystems.

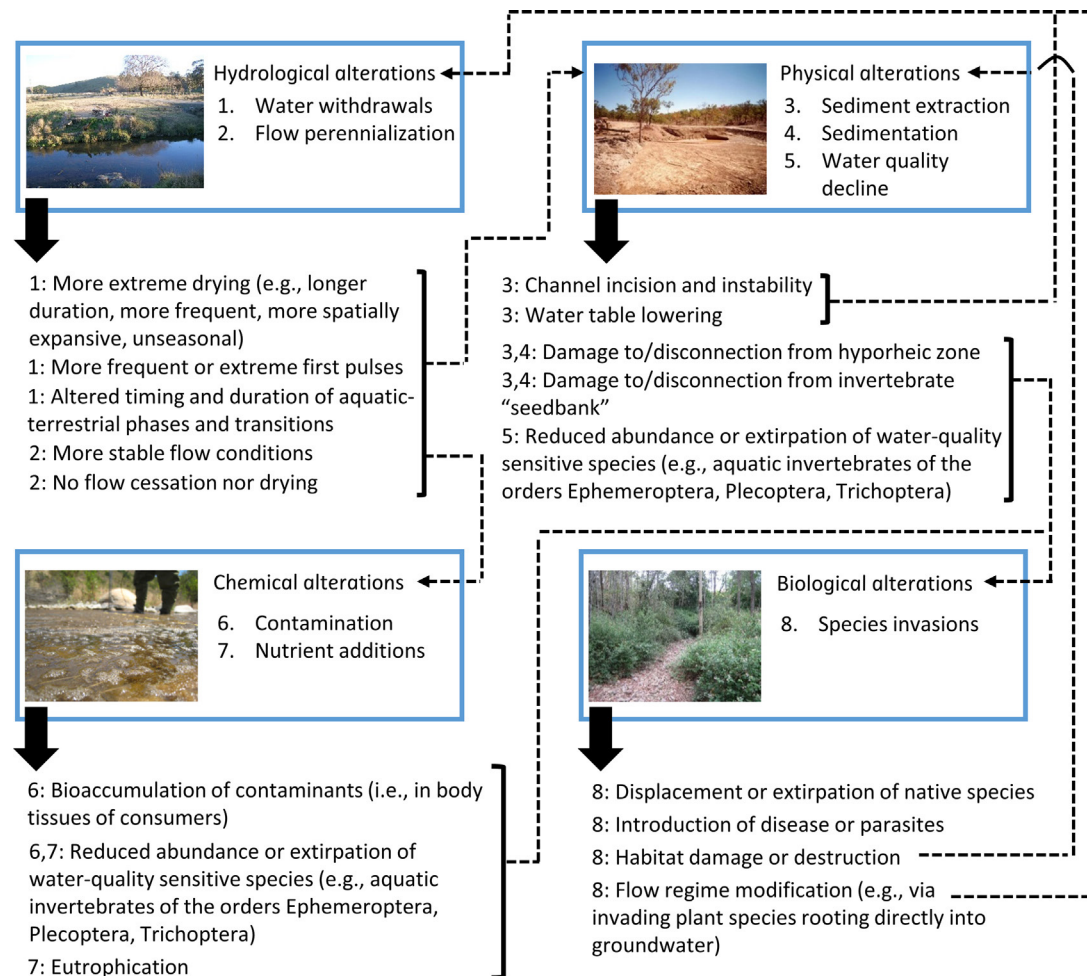
5.1.2 HYDROLOGICAL ALTERATIONS IN IRES

Flow regulation, surface and groundwater abstraction, and water diversions for flood control, irrigation, and other purposes have altered the flow regimes of many rivers worldwide (Vörösmarty and Sahagian, 2000). Anthropogenic hydrological alterations in rivers are ubiquitous (Dudgeon et al., 2006), especially in regions where flow regimes are highly seasonal or variable, and aim to meet human demands for

FIG. 5.1.1

Examples of anthropogenic impacts that affect or create IRES: (a) an artificial IRES created by a small hydropower station in Riu Freser, Girona, Catalonia, Spain; (b) eutrophic conditions of San Juan Creek, an IRES in Orange County, California; (c) a heavily modified IRES (rambla) used for vehicular transport in Río Seco, Málaga, Spain; (d) mining activities within and surrounding Sandy Creek, an IRES in northeast Australia; (e) an IRES below a dam built by an aggregate-extraction company in Barranc de l'Estany, Tarragona, Spain; (f) invasion of *Lantana camara* throughout the riparian zone of a forested headwater IRES in Brisbane, Queensland, Australia, traversed by a well-used bush-walking track; (g) nonnative Mexican fan palms (*Washingtonia robusta*) are frequent invaders in IRES experiencing hydrological alteration in southern California: a young sapling has taken root in Jeronimo Creek (Orange County, California), adjacent to a stormwater outfall that has perennialized the creek's lower portions.

Photos: Courtesy P. Bonada (a), R. Mazor (b and g), N. Bonada (c), F. Barron (d), N. Cid/TRivers project (e) and C. Leigh (f).

**FIG. 5.1.2**

The prevailing anthropogenic threats to IRES (blue-lined boxes) and interactions among them: hydrological (water withdrawals; perennialization), physical (sediment extraction; sedimentation; physical water quality decline such as hypoxia), chemical (contamination by heavy metals and other toxic substances; nutrient additions), and biological (species invasions) alterations. Thick arrows indicate potential consequences of threats, as numbered within boxes. Thin broken lines with arrowheads show where potential consequences of threats may exacerbate existing states or produce novel changes in the hydrological, physical, chemical, or biological conditions of IRES. See text for more details.

Photos: Courtesy R. Rolls (hydrological alterations; water abstraction from a small IRES in northern New South Wales, Australia), F. Barron (physical alterations; sediment mining in northeast Australian IRES), C. Leigh (biological alterations; Lantana camara invasion of an eastern Australian IRES), R. Mazor (chemical alterations; eutrophication of an IRES in western United States).

BOX 5.1.1 CLIMATE CHANGE AND IRES

IRES are especially sensitive to climate change because the durations of their wet and dry phases, and the thermal thresholds of the organisms present, are strongly affected by precipitation patterns and temperature maxima (Brooks, 2009). Ensuing hydrological changes such as increased flow intermittence (e.g., Döll and Schmied, 2012; van Vliet et al., 2013) could have similar ecological impacts (e.g., species extinctions, altered biogeochemical processes; Jaeger et al., 2014) on IRES to those of direct anthropogenic manipulations of flow involving water removal (e.g., Dewson et al., 2007).

In the dry phase of IRES, climate change may result in local extinctions, species replacement, and alteration of metapopulation dynamics (Chapter 4.9) by increasing flow intermittence and reducing habitat connectivity and availability (Bond et al., 2010; Jackson and Sax, 2010), as well as leading to changes in accumulations of organic matter on dry streambeds (Boulton and Lake, 1992; Section 5.1.5). Biogeochemical processes will be influenced by extended drying (Chapter 3.2) and the increased possibility of extreme flooding from increased precipitation (Acuña and Tockner, 2010).

Increased water temperature could have similar effects to the impacts of many anthropogenic activities (Section 5.1.3). In addition, future warming may increase instream metabolism of organic carbon in IRES, but these effects are reported to be much smaller than those caused by flow regime alterations (Acuña and Tockner, 2010). Considerable uncertainty still surrounds the likely impacts of current threats superimposed on the effects of climate change in IRES, especially where stressors are likely to combine in a nonadditive way and provide novel conditions that exceed the tolerances of native biota.

regular, reliable, and safe water supply and storage (Grey and Sadoff, 2007). Unfortunately, this means that IRES are often under threat of or subject to anthropogenic flow alteration. IRES are renowned for their flow variability (Davies et al., 1994; Puckridge et al., 1998; Poff et al., 2006; Chapter 2.2), not least because of their continual alternations between the two extremes of flow—flooding and drying—that create complex mosaics of dry channels and lentic waters interspersed among flowing reaches and floodplains (e.g., Gallart et al., 2012; Datry et al., 2016; Chapters 1 and 2.3). The importance of this natural flow variability to the ecology of river systems, including IRES, was recognized many years ago (e.g., Walker et al., 1995; Poff and Ward, 1989). Accordingly, scientists raised concerns about the threats posed by anthropogenic hydrological alterations to the ecological integrity of IRES (e.g., Walker et al., 1997; Boulton et al., 2000); these are concerns that remain to this day as discussed later.

HYDROLOGICAL ALTERATIONS ASSOCIATED WITH ANTHROPOGENIC WATER WITHDRAWALS

Water removed from rivers and aquifers by abstraction, retention, or diversion for exploitation by humans (hereafter, “water withdrawals”) can alter the hydraulics and hydrology of rivers. In perennial streams and rivers, water withdrawals may cause flow cessation and drying when and where these events would not otherwise occur (e.g., exposing dry channel beds in reaches otherwise permanently watered, Benejam et al., 2010). This may also occur as a result of water withdrawals from natural IRES, along with prolonged periods of drying (King et al., 2015; Table 5.1.1). Changes in the timing, location, duration, and frequency of flow cessation and drying in natural IRES or novel flow cessation and drying in once-perennial rivers may place biota at risk of local or regional extinction (Box 5.1.1). For example, in the once-perennial Tordera River in Spain, from which 34% of its water is abstracted for human use, the channel is dry 97% of the time when it would otherwise flow and native fish species have consequently declined in number (Benejam et al., 2010). Water abstraction from IRES for irrigation is also linked to the proliferation of nonnative fish species (Godinho et al., 2014), which may then

Table 5.1.1 Hydrological alterations resulting from anthropogenic water withdrawals, specifically those inducing flow cessation or drying (From King, A.J., Townsend, S.A., Douglas, M.M., Kennard, M.J., 2015. Implications of water extraction on the low-flow hydrology and ecology of tropical savannah rivers: an appraisal for northern Australia. *Freshw. Sci.* 34, 741–758.), with examples of observed biotic changes that may ensue

Natural flow regime	Altered flow regime	Hydrological alteration and habitat changes	Biotic changes
Intermittent/ephemeral	Intermittent/ephemeral	Lower volumes during period of flow Earlier, longer and more spatially extensive periods of flow cessation or drying, and possible novel flow cessation or drying Shorter persistence of isolated pool habitats	Changes as per those documented for perennial rivers, below Rapid recovery of macroinvertebrate communities to pre-drying composition if source(s) of colonists remain intact (Vander Vorste et al., 2016) Potential collapse of freshwater food webs (Chapter 4.7) and slowed recovery of biotic communities following flow resumption
Perennial	Intermittent/ephemeral	Lower flow volumes Novel flow cessation, isolated pool habitats and/or drying Novel flow resumption and post-drying period	Increase in density of macroinvertebrates associated with decreased habitat area (Dewson et al., 2007) Decline in proportion of intolerant fish species (e.g., those showing restricted distributions or decrease in abundances in response to anthropogenic disturbances) and benthic fish species (Benejam et al., 2010); Potentially no change in macroinvertebrate diversity but change in composition, e.g., proportional increase in Chironomidae taxa (Hille et al., 2014); Populations of fishes and other obligate lotic species eliminated during drying Rapid recovery of macroinvertebrate communities to pre-drying composition if source(s) of colonists remain intact (Dudgeon, 1992); Partial or slow recovery of fish populations if source(s) of colonists remain intact (Skoulikidis et al., 2011)

place additional pressure on the native taxa (Section 5.1.4). On a regional scale, for example, the IUCN considers water abstraction one of the greatest threats to fishes of the Mediterranean basin (Smith and Darwall, 2006), a well-recognized hot spot of freshwater biodiversity (Myers et al., 2000) where many rivers undergo natural or anthropogenic flow intermittence.

The effects of anthropogenic water withdrawal on river habitat, water quality, and biota appear to mimic those described for meteorological drought (Lake, 2011) as well as natural drying (Boulton, 2003); aquatic habitats contract and become fragmented, along with declines in water quality and changes to food resource availability (Rolls et al., 2012; Section 5.1.3). Consequently, the survival, reproduction, and movement of water-dependent species are typically reduced. With complete loss of

surface water, aquatic biota are eliminated from surface habitat, but they can recolonize following flow resumption (Kawanishi et al., 2013; Vander Vorste et al., 2016).

However, the rate and success of recolonization depends on the proximity and availability of aquatic refuges, the intensity and duration of drying, the traits of the taxa involved (Chapter 4.8) and the cause of intermittence (natural vs anthropogenic). In once-perennial rivers, for example, recovery of fish communities following extensive channel desiccation resulting from water abstraction may take several years of “normal” streamflow (Skoulikidis et al., 2011), whereas macroinvertebrate communities may recover rapidly after flow resumption as long as upstream sources of colonists remain intact (Dudgeon, 1992; Skoulikidis et al., 2011). However, when water quality declines (e.g., increased conductivity and temperature, decreased dissolved oxygen; Section 5.1.3) in tandem with water volume, the recovery of macroinvertebrate communities may be delayed and there may be extreme changes in community composition (Box 5.1.1). For example, temperatures exceeding 30°C, together with water abstractions and diversions for irrigation extracting over 90% of original discharge volumes and causing instream drying, shifted macroinvertebrate community composition from dominance by detritus-collecting Ephemeroptera, Plecoptera, and Trichoptera taxa to dominance by predatory insects, noninsect taxa, and algae-scraping beetles (Miller et al., 2007). In summary, the persistence of aquatic biota is threatened by persistent and intense water withdrawals that induce or prolong river drying.

HYDROLOGICAL ALTERATIONS THAT INDUCE PERENNIAL FLOW IN NATURAL IRES

Flow regulation and the addition of water to IRES can induce perennial flow in natural IRES, a process we refer to as “perennialization” and for which the causes are manifold. For example, IRES channels can be used to convey water for livestock and irrigation or other uses, particularly during dry seasons when flow would naturally cease (Gasith and Resh, 1999; Bond et al., 2010). In urban areas, impervious surfaces and landscape irrigation with imported water (Roy et al., 2009; Wenger et al., 2009) or releases of wastewater effluent (Bischel et al., 2013; Lawrence et al., 2014) can cause IRES to flow continuously. Nutrient-rich wastewater inputs to ephemeral streams of Israel and Palestine, for example, have created continuous flow and caused the rapid development of riparian vegetation (Hassan and Egozi, 2001). Removal of riparian vegetation and consequent reduction in evapotranspiration may actually induce perennialization (Ingebo, 1971; Salemi et al., 2012). For example, Hibbs et al. (2012) observed that perennialization in southern California resulted from a combination of factors, including vegetation removal, importation of water, and increased imperviousness, that together incised stream channels below the groundwater table.

Although perennialization of individual reaches has been documented for some IRES (e.g., Bond et al., 2010; Hibbs et al., 2012), the spatial extent of this phenomenon across river networks and at the global level is unknown and remains poorly understood. Stream surveys at the local level, however, have found that systems in urban and agricultural areas have higher proportions of perennial stream length than expected, relative to those in comparable natural areas, a difference attributed to anthropogenically augmented flows from discharge and runoff (SWAMP, 2011; Mazor, 2015a).

When considered as “streamflow augmentation,” perennialization is commonly associated with benefits such as increased water quality, habitat area for aquatic and riparian biota, and esthetic and recreational amenity (Brooks et al., 2006). Calera Creek, an urban IRES in California that has been perennialized by tertiary-treated effluent discharged in unnatural diurnal pulses, now has “improved”

water quality relative to that of its highly degraded, nonperennialized reaches upstream (Halaburka et al., 2013). Other benefits include a drastic reduction in pest mosquito populations and an overall increase in benthic macroinvertebrate abundance relative to the degraded, nonperennialized reach; however, diversity has remained relatively low (Halaburka et al., 2013). The augmentation project also involved creation of an urban park and extensive aquatic habitat restoration. Four hectares of wetland and about 20 ha of stream bank and buffer areas were revegetated, habitat was recreated for the endangered San Francisco garter snake (*Thamnophis sirtalis tetrataenia*) and the threatened Californian red-legged frog (*Rana draytonii*), and the formerly diverted and channelized lower creek was returned to its historic flowpath. Thus, the physical, biological, and social benefits of the project cannot be attributed exclusively to perennialization and research is needed to identify the ecological effects (“positive” or “negative”) of perennialization on this and other IRES.

Legitimate concerns have been raised about perennialization of IRES (Brooks et al., 2006). For example, both public and ecosystem health may be at risk depending on the quality of the water perennializing the IRES (Section 5.1.3), which might contain pharmaceutical products, trace contaminants or elevated nutrient levels, or be released at temperatures dramatically different to that of the receiving waterbody (e.g., treated effluent has been recorded at temperatures up to 10°C above that of receiving stream water; Plumlee et al., 2012). Restoring a natural hydrograph (i.e., restoring periods of flow cessation and drying) to perennialized reaches of IRES may even facilitate the recovery of native assemblages. For example, mathematical modeling of fish distributions in the Broken-Boosey Creek system, a southeast Australian IRES which has been perennialized by flow regulation for over 100 years, predicts that flow restoration will facilitate the reestablishment of populations of at least two native fish species that have long been eliminated from the system (Bond et al., 2010).

5.1.3 PHYSICAL AND CHEMICAL ALTERATIONS IN IRES EXPLOITATION OF IRES AND DRY RIVERBEDS BY HUMANS

Dry riverbeds of IRES are exploited by humans for numerous uses such as recreation and resource extraction (Steward et al., 2012). The sediment of dry channels is exposed and, depending on the IRES location and surrounding infrastructure, constitutes a readily accessible commodity for mining. This is often done using heavy machinery such as bulldozers which extract sand and gravel directly from dry riverbeds, a process known as dry-pit mining (Kondolf, 1994). The physical and ecological effects of such intense activity are diverse: riparian and instream vegetation is damaged and sediment deficits can cause lateral channel instability, bed armoring, and progressive incision both upstream and downstream (e.g., headcutting) that may then alter the frequency of floodplain inundation, lower the water table, and even destroy infrastructure such as bridges (Rinaldi et al., 2005). In the Upper Ewaso Ngiro North River Basin of Kenya, for example, sediment-mining activities are estimated to have lowered the ephemeral riverbed by up to 2 m (Gichuki, 2004).

Removal of the overlying layer of moist and even dry sediment during dry-pit mining is likely to negatively affect the recolonization processes that occur in IRES upon flow resumption and habitat rewetting (Chapter 4.8). When waters recede in anthropogenically unmodified IRES, moist sediments in the surface and potentially the subsurface (e.g., the hyporheic zone) are used as refuges by many animals, including invertebrates and, in rare cases, fish that can then recolonize the surface habitat when flow returns (Kawanishi et al., 2013; Vander Vorste et al., 2016). These moist zones provide a potential

refuge that promotes the survival of biota lacking life history adaptations to withstand complete drying (Stubbington, 2012; Chapter 4.3). The macroinvertebrate “seedbank” within sediments of dry riverbeds can also contribute to the recolonization and persistence of aquatic communities in many IRES (Storey and Quinn, 2008; Datry, 2012; Stubbington and Datry, 2013).

Dry riverbeds can also provide a fertile but often scarce source of land for short-term agricultural activity (e.g., Hans et al., 1999) or use as roads (Fig. 5.1.1), which can increase both the mobilization and compaction of river sediments (e.g., by releasing fine sediments that clog substrate interstices). This forms a physical barrier that restricts movement of biota into the hyporheic zone or “seedbank,” limiting the functions of these refuges during surface water loss (Vadher et al., 2015). This refuge function is also threatened by livestock grazing, which in IRES and perennial rivers alike can cause bank erosion, increase soil compaction and nutrient concentrations, and remove or damage riparian vegetation (Belsky et al., 1999; Herbst et al., 2012). Even light trampling by livestock can cause compaction and destroy habitat features of dry riverbeds such as biological or salt crusts and mud cracks or curls (e.g., Steward et al., 2012; CWMW, 2015; Chapter 4.4) that support unique assemblages of terrestrial invertebrates, identified as distinct from those inhabiting adjacent upland areas (Steward et al., 2011; Chapter 4.4).

Such effects are not limited to livestock; activities of feral animals may have similar impacts on IRES sediments and habitat. For example, Doupe et al. (2009) documented the degradation of ephemeral wetlands on the floodplain of Laura River, an IRES in tropical north Australia, by feral pigs (*Sus scrofa*) whose waste excretions and foraging activities combined to destroy macrophyte communities, upheave sediments, increase turbidity by several orders of magnitude and produce anoxic conditions, pH imbalances, and nutrient enrichment that persisted for months.

IRES AND WATER QUALITY DEGRADATION

Inputs of contaminated water, excess nutrients, and sediment from anthropogenic activities threaten the water quality of rivers the world over, including IRES. In IRES, inputs that occur during flow cessation or drying are not diluted by river discharge and usually remain *in situ* until flow resumes. Alternatively, if inputs occur during flow, the contaminants, nutrients, or sediments may be transported and deposited in downstream accumulations when flow ceases. In either case, flow intermittence may prolong the exposure of riverine and riparian biota to contaminants, nutrients, and sediments for longer and at higher concentrations than might occur if flow was continuous (e.g., Resh and Jackson, 1993). In the unregulated Guadiamar River, an IRES in southwest Spain that dries completely in several reaches during summer, a major spill from a zinc and silver mine during the spring of 1998 resulted in 5 million cubic meters of toxic waste entering the system (Sola et al., 2004). Even after 2 years of restoration activities involving removal of metal-polluted sediment, fewer than half the macroinvertebrate taxa present in surrounding unaffected reaches were found in the impacted reach, whose sediments still had concentrations of heavy metals up to 120-fold higher than control concentrations (Sola et al., 2004). Metal concentrations in *Hydropsyche* caddisflies collected from the impacted reach were up to 35 times higher than those at control sites (Sola et al., 2004).

Inputs of excess nutrients and pharmaceuticals to IRES from wastewater effluents can also lead to water quality impairments such as hypoxia or elevated pH, cause eutrophication, and expose biota directly to harmful chemicals (Dodds and Welch, 2000; Camargo and Alonso, 2006). For example, the upper Arc River in Provence, France, which naturally ceases flow in several reaches during summer, receives inputs of treated and sometimes untreated sewage from wastewater treatment plants.

Therefore, its flow during this season is predominantly effluent (Comoretto and Chiron, 2005) and is also when the highest concentrations of the pharmaceuticals carbamazepine ($1.15 \mu\text{g L}^{-1}$) and bezafibrate ($0.78 \mu\text{g L}^{-1}$) are observed, exposing multiple generations of the river's aquatic inhabitants to these pollutants (Comoretto and Chiron, 2005).

Although few studies provide direct comparisons, the hydrology of IRES probably render them more sensitive to nutrients and other pollutants than comparably sized perennial rivers. Indeed, the risks of eutrophication are greater in the low-velocity, warm conditions that are typical of IRES during low-flow conditions and flow cessation when pools of surface water become isolated, as seen for example in the Evrotas River in Greece (Tzoraki et al., 2014). In IRES of southern California, indicators of eutrophication such as streambed-smothering growth of filamentous green *Cladophora* sp. and large fluctuations in pH and dissolved oxygen have been observed, even in the IRES with moderate nutrient concentrations (e.g., $<1 \text{ mg L}^{-1}$ total nitrogen; Mazor et al., 2012; Mazor et al., 2014). These IRES also receive poor biological integrity scores, suggesting that small additions of nutrients may have disproportionately large impacts on the water quality and biota of IRES (Mazor et al., 2012; Mazor et al., 2014). Similar sensitivity has been noted in the intermittent Fuirosos River in Spain, where moderate nutrient inputs (i.e., twofold and threefold increases in nitrogen and phosphorus, respectively, above ambient concentrations) shifted the taxonomic and trophic structure of the biota by producing conditions favoring organisms such as grazing snails that could take advantage of the resultant increases in algal resources and fine particulate organic matter (Sabater et al., 2011).

5.1.4 BIOLOGICAL ALTERATIONS IN IRES: INVASIVE SPECIES

Invasion of fresh waters by alien species (Fig. 5.1.1) is one of the five major threats to freshwater biodiversity (Dudgeon et al., 2006) and this is no less true for the biodiversity of IRES. Invasive species can prey upon or compete with native biota for resources and destroy native species' habitat. Because native biota can be constrained to low population densities under harsh conditions (Chesson and Huntly, 1997), such as can occur in IRES during flow cessation and drying (e.g., Fritz and Dodds, 2005; Chapter 3.1), they may be highly susceptible to extinction from predation by or competition with invasive species. For example, the recruitment and persistence of low-density populations of the native Sonora mud turtle (*Kinosternon sonoriense*) which inhabits ephemeral Arizonan IRES are threatened by the invasion of the alien crayfish (*Orconectes virilis*) because it preys on the hatchlings of this vulnerable species (Hensley et al., 2010).

Invasive species also threaten IRES via processes or behaviors that modify flow regimes or habitat. For example, the deep-rooted riparian tamarisk (*Tamarix* spp.), an alien species in the United States and Australia, grows quickly and can tolerate long periods of drought and inundation, making it well suited to colonization of riverbanks, especially along IRES (Tickner et al., 2001; Stromberg et al., 2007). Deep alluvial groundwater below IRES favors tamarisk over the shallower-rooted, native riparian plants (e.g., Stromberg et al., 2007). Because tamarisk can root directly into groundwater, it can contribute to flow cessation and river drying through significant alteration of groundwater hydrology. Once dense stands are established, their trapping and stabilization of sediment can alter fluvial geomorphology and instream habitat (Tickner et al., 2001). Feral animals and livestock that concentrate around permanent pools in IRES channels following flow cessation can trample shallow habitat, contaminate waters with feces and urine, and increase water turbidity, potentially shifting primary producer communities from dominance by aquatic macrophytes to dominance by cyanobacteria (Pettit et al., 2012).

Alien species can also introduce parasites into IRES, threatening native fauna previously unexposed (Clavero et al., 2015). For example, Stone et al. (2007) found that nonnative fishes provided a mechanism for the dispersal of alien aquatic species in IRES; species such as common carp (*Cyprinus carpio*) that hosted the parasitic Asian tapeworm (*Bothriocephalus acheilognathi*) were potential invaders of habitat along the Little Colorado River, Arizona, having survived in upstream pools over the dry period and moved downstream with subsequent freshets. Indeed, some nonnative species appear to be favored by or can at least tolerate IRES flow regimes. For example, extended drying in Californian IRES facilitates establishment and success of the invasive green sunfish (*Lepomis cyanellus*) which also tolerate the extreme high flows that occur in these systems (Bêche et al., 2009; Fig. 5.1.3).

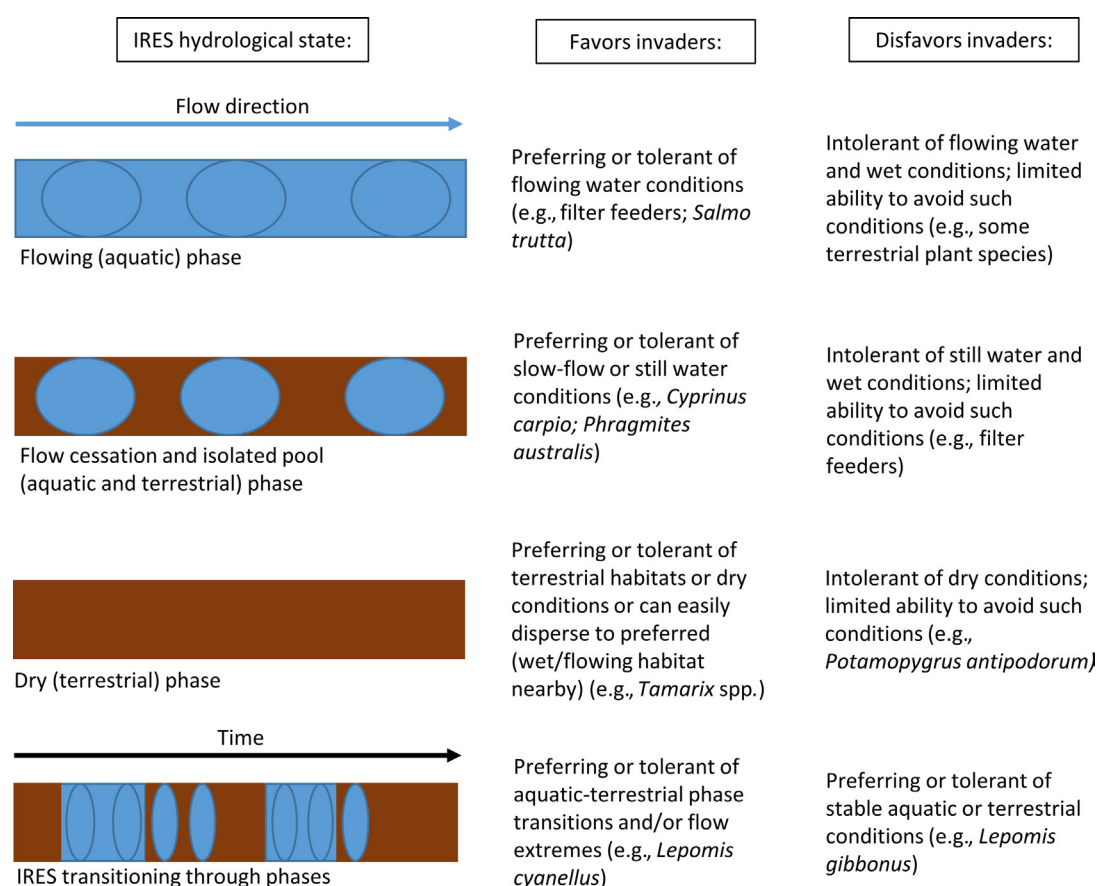


FIG. 5.1.3

Hydrological conditions of IRES favoring or disfavoring the invasion of alien species as dependent on species tolerances, intolerances or dispersal abilities, with some examples provided (for more detail, see text and Closs and Lake, 1996; Tickner et al., 2001; Koehn, 2004; Stone et al., 2007; Abramson, 2009; Arce et al., 2009; Bêche et al., 2009; Clavero et al., 2015).

Conversely, the natural flow variability of IRES may offer some protection against invasion by alien species (Costelloe et al., 2010; Fig. 5.1.3), because many alien species depend on permanent aquatic habitats. In an Australian IRES, the Lerderderg River, nonnative *Salmo trutta* had high levels of mortality during an extended dry period whereas native *Galaxias olidus* survived in isolated pools, suggesting that the nonnative species was constrained by its inability to tolerate the naturally harsh conditions that occurred during summer flow cessation and subsequent habitat fragmentation (Closs and Lake, 1996). Likewise, unregulated upstream reaches of Moroccan IRES are presently relatively free of pumpkinseed sunfish (*Lepomis gibbosus*) and other nonnative fishes, but these species appear to have successfully colonized the flow-regulated reaches downstream and consequently lowered the abundance of native fish (Clavero et al., 2015). In California, many of the most widespread invasive animals (e.g., bullfrogs *Lithobates catesbeinus*, New Zealand mudsnails *Potamopyrgus antipodarum*, mosquito fish *Gambusia affinis*) cannot tolerate lengthy dry periods. For example, in Malibu Creek in California, mudsnails have invaded nearly all perennial reaches, which means that intermittent reaches (and, in some cases, upstream perennial reaches) now remain the only uninfested portions of the watershed (Abramson, 2009). Thus, intermittent reaches may provide critical refugia from alien species, and protecting these refugia may be essential to sustaining the overall biodiversity of the watershed.

5.1.5 CASCADING EFFECTS OF HYDROLOGICAL, PHYSICAL, CHEMICAL, AND BIOLOGICAL ALTERATIONS IN IRES ON ADJACENT ECOSYSTEMS

The natural flow regimes and the presence and prevalence of IRES can strongly influence downstream ecosystems by altering the timing, quality, and quantity of sediment, water, and nutrients delivered to them (Chapters 3.1 and 3.2). During drying, large volumes of riparian leaf litter and sediments may accumulate, and terrestrial vegetation gradually replaces aquatic plants and algae in the dry riverbeds (Steward et al., 2012; Datry et al., 2014). When stream flow resumes, these materials are released and transported downstream by advancing wetted fronts (Corti and Datry, 2012) which rewet dry riverbeds and entrain terrestrial and aquatic invertebrates, suspended sediments, and both fine and particulate organic matter. For example, in the Albarine River in France, invertebrates in an advancing wetted front were found to be mainly of terrestrial origin, and entrained aquatic taxa represented ~15% of the benthic taxa downstream (Corti and Datry, 2012).

Hydrological alterations can influence these material fluxes and cause cascading ecological effects on adjacent ecosystems, including downstream aquatic and terrestrial habitats. For example, hypoxic blackwater events in IRES can occur when large amounts of organic material accumulated during natural or anthropogenic drying are mobilized and metabolized following flow resumption. These blackwater events cause dramatic declines in dissolved oxygen, killing fish and zooplankton in downstream waters (King et al., 2012; Small et al., 2014). The likelihood of these events can depend on the amount and type of accumulated carbon in the dry channel, the duration and spatial extent of the preceding zero-flow period, and the temperature and timing of rewetting flow (Acuña and Tockner, 2010; Hladysz et al., 2011; Datry et al., 2014). Prolonged drying (i.e., beyond that which would occur naturally) caused by either climate change (Box 5.1.1) or anthropogenic activities such as water withdrawals can result in greater quantities of organic material accumulating. Anthropogenic activities can also add directly to these accumulations (e.g., forestry slash entering streams, leaf fall after intentional fire).

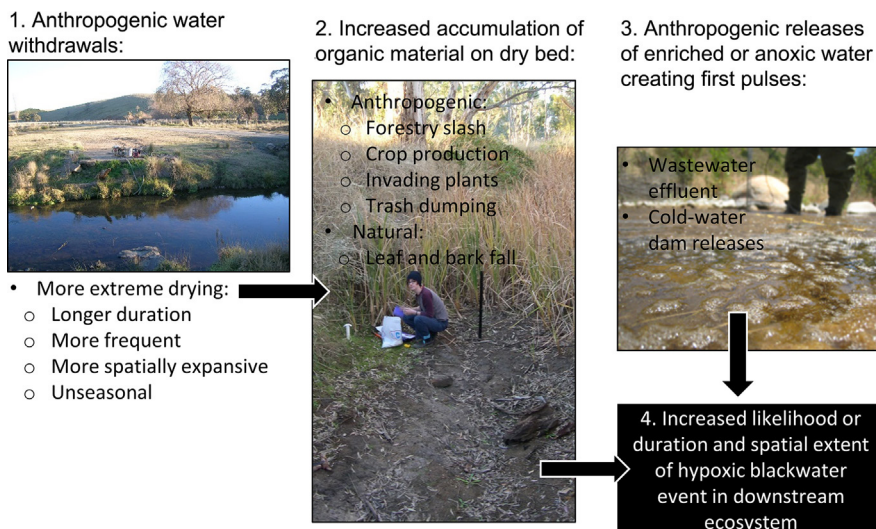


FIG. 5.1.4

Cascading and interactive effects of anthropogenic hydrological and chemical alterations in IRES (as depicted in Fig. 5.1.2) and potential consequences for downstream ecosystems.

Photos: Courtesy R. Rolls (anthropogenic water withdrawals from a small IRES in northern New South Wales, Australia), F. Sheldon (Catherine Leigh observing the organic matter accumulated on the dry bed of an IRES near Adelaide, Australia), R. Mazor (anthropogenic eutrophication of an IRES in California, United States). See text for more details.

Consequently, the likelihood of a hypoxic event occurring upon flow resumption increases. Higher magnitude and/or unseasonal flushing flows from effluent inputs or dam releases onto dry riverbeds could further increase the potential occurrence, extent, and duration of such events in IRES and other ecosystems downstream (Fig. 5.1.4).

Extended drying or perennialization in IRES may also have cascading effects on riparian ecosystems. Many riparian animals such as arthropods, bats, lizards, and birds rely on consumption of aquatic biota from IRES, particularly during nonflow periods when aquatic biota are concentrated in isolated pools (e.g., Leigh et al., 2013a). Hydrological alterations that cause these pools to dry early (Table 5.1.1) could thus reduce the longevity of aquatic food supply for consumers inhabiting the riparian zones of IRES. Vegetation in riparian zones may also be affected by hydrological alterations in IRES because water availability is a key driver of distribution patterns of riparian plant species (Stromberg and Merritt, 2016). For example, normalized difference vegetation index scores have shown that water loads from anthropogenic sources in southern California have not only perennialized IRES but also supplemented runoff to riparian habitats during naturally dry phases (Cervantes, 2013).

Anthropogenic alterations of flow regimes, water quality, and sediment dynamics in IRES could have cascading effects not only on neighboring (perennial, intermittent, or ephemeral) reaches and riparian zones but also on hyporheic zones, as discussed in Section 5.1.3. Beyond the hyporheic zone, first pulses in IRES can remobilize sediments with adsorbed pollutants and carry them downstream

or deposit them on riverbanks and floodplains (Obermann et al., 2009). For example, in the Vène River in southern France, which receives inputs of wastewater effluent into its intermittent reaches, accumulation of anthropogenic pollutants (organotins, fecal indicator bacteria, and nutrients) in a downstream lagoon that accounts for 10% of French shellfish production occurs predominantly via sediment resuspension and pollutant remobilization during high flows (Chahinian et al., 2012). With runoff and sediment mobilization occurring only 10–15 s after rainfall starts, dry riverbed sediments are also an important source of nutrients and bacterial contamination in waters downstream from IRES, particularly in agricultural watersheds. This runoff poses risks not only to water quality and river biota but also to human health (Frey et al., 2015).

In terms of biological alterations, alien species that occupy IRES can also invade adjacent ecosystems. For example, in the Little Colorado River in Arizona, nonnative fishes such as *C. carpio* are restricted to isolated pools after flows cease, but they invade downstream perennial reaches during flash floods, threatening local native fish populations (e.g., endangered humpback chub, *Gila cypha*) through resource competition and infection by introduced parasites such as *B. acheilognathi* as discussed in Section 5.1.4 (Stone et al., 2007).

The anthropogenic hydrological, physical, chemical, and biological alterations that threaten IRES also threaten multiple adjacent ecosystems, often in complex and interactive ways (Fig. 5.1.4). Although the threats to IRES are, at some level, similar to those facing or that can cascade into perennial streams and rivers, there are some important differences. IRES are subjected to certain threats rarely experienced by perennial rivers (e.g., exploitation of the dry riverbed) and respond to other threats more acutely (e.g., sensitivity to nutrients and pulsed contaminants). IRES contribute significantly to regional biodiversity (e.g., their unique terrestrial invertebrate fauna) and play outsize roles in certain ecosystem services such as processing of nutrients and organic matter (Chapter 3.2). Effective management and condition assessment of aquatic resources in IRES are incomplete without consideration of these differences.

5.1.6 THE CHALLENGE OF ASSESSING THE BIOLOGICAL AND ECOLOGICAL CONDITION OF IRES

The management of threats and mitigation of their impacts on IRES requires the implementation of integrated assessments within monitoring programs that explicitly consider hydrological variability (Bond and Cottingham, 2008; Prat et al., 2014; Box 5.1.2; Chapter 5.5). Biological and ecological assessments are widely used in streams and rivers worldwide (Carter et al., 2006) but are still in development for IRES (Nikolaidis et al., 2013; Mazor et al., 2014; Leigh et al., 2016b). Many regional surveys of stream condition explicitly or effectively exclude IRES (e.g., Hall et al., 1998; Sheldon, 2005; Dobbie and Negus, 2013), often because information on the spatial and temporal hydrological conditions of the river is missing (i.e., when, where, and for how long does the IRES flow?). This makes it difficult or impossible for managers to use existing standard methods developed for perennial streams and rivers to assess IRES.

In regions where IRES prevail, biological and water samples may not be collected if the target reach is dry at the time of sampling. Furthermore, if samples are collected during flow or when disconnected pools are present, biological quality may not effectively represent the river's ecological condition (Sheldon, 2005; Munné and Prat, 2011; Mazor et al., 2014). For instance, biotic indices from “reference” and “least

BOX 5.1.2 IMPROVING BIOASSESSMENT OF IRES: TWO EXAMPLES

The MIRAGE toolbox (Prat et al., 2014), now incorporated into the open access software TREHS (Temporary Rivers Ecological and Hydrological Status), is a new approach developed to improve assessment of IRES in the Mediterranean basin. The toolbox allows users to classify a river's flow regime as "perennial," "intermittent-pools," "intermittent-dry," and "ephemeral/episodic" and provides information on the frequency of different aquatic states (e.g., flowing, isolated pools, dry) occurring throughout the year (Gallart et al., 2012; Table 5.1.2). It integrates data from several sources, including flow data from gauging stations, modeling, interviews, and aerial photographs (Gallart et al., 2016) and simultaneously conducts a hydrological assessment to ascertain whether flow cessation is due to natural or anthropogenic causes. This information then facilitates decisions on the potential application of existing standard methods of ecological assessment for each river (Table 5.1.2).

Demonstration of the validity of bioassessment indices in IRES is crucial to including them in management programs. For example, a recently promulgated "total maximum daily load" in Malibu Creek, California, demonstrated that poor bioassessment scores were associated with excess nutrients and sedimentation, not with the intermittent flow regimes prevalent in the watershed (USEPA, 2013). In arid southern California, an ongoing probabilistic regional stream monitoring program recently expanded its target population to include IRES (Mazor, 2015b) and will assess them with benthic macroinvertebrate and algal indices, along with riparian wetland assessment indicators (e.g., CWMW, 2015). Indicators to assess highly ephemeral portions are in development, as are predictive models to estimate flow regimes at ungauged sites. Data from these regional assessments will be used to include IRES in management programs intended to protect high-quality streams, prevent degradation relative to baseline conditions, and prioritize degraded streams for improvement.

Table 5.1.2 Potential use of existing standard methods to assess biological and ecological condition of natural IRES

Natural flow regime as defined within the MIRAGE toolbox	Can current standard biotic indices developed for perennial streams and rivers be used
Intermittent-pools: Flow cessation occurs during the dry season maintaining disconnected pools	Yes. Sampling to be conducted during flow. Sampling calendar needs to be adapted for each station and conducted in those months in which the probability of finding flow is the highest
Intermittent-dry: Flow cessation occurs during the dry season and the IRES dries completely at its surface	No. Biological indices based on aquatic fauna need to be adapted and sampling needs to be conducted during flow. Development of new methods, not based on aquatic fauna, could be an option (Chapter 4.4)
Ephemeral/episodic: Flow and disconnected pools are occasional	No. Development of new methods, not based on aquatic fauna, is required

disturbed" sites (used as standards against which the conditions of other streams are assessed) in the Mediterranean basin were so variable, even when they were only sampled while flowing, that defining environmental targets for this region's IRES became extremely difficult (Sanchez-Montoya et al., 2010; Feio et al., 2014). This difficulty is partly due to the responses of the aquatic biota inhabiting IRES to flow cessation and drying, particularly invertebrates which are commonly used as biological indicators. As discussed earlier, when surface-water pools disconnect following flow cessation, the number of invertebrate taxa usually declines (Boulton, 2003) and those adapted to flow cessation (e.g., many Odonata, Coleoptera and Heteroptera: OCH) become predominant (Bonada et al., 2006; Cid et al., 2016). This reduced taxonomic richness, together with the low values that OCH taxa often score in many biological assessment schemes and indices (Chessman, 2003; Alba Tercedor et al., 2004), essentially forces managers to assess IRES only during times of flow (Bonada et al., 2006; Buffagni et al., 2010).

The use of terrestrial or hyporheic invertebrates as alternative indicators of biological condition during dry or zero-flow periods is now being explored for IRES (e.g., Steward et al., 2012; Corti et al., 2013; Leigh et al., 2013b) and concerted efforts are leading to improvements or new practices and tools for assessing the ecological condition of IRES (Box 5.1.2). In California, for example, assessment indices based on benthic macroinvertebrates (e.g., Ode et al., 2005) or algae (e.g., Fetscher et al., 2014) have been calibrated with nominally perennial reference streams. Most of these streams exhibit dynamic flow regimes and many are actually IRES, particularly in dry years. Consequently, the indices perform well in some IRES and show potential for setting management targets for biological integrity (Mazor et al., 2014).

5.1.7 CONCLUSIONS

IRES commonly occur throughout the world, but the ecosystem services that they provide are often unappreciated (Chapter 5.2). In addition to the same hydrological, physiochemical, and biological impacts that adversely influence perennial systems, IRES are subject to specific types of impact when they cease flow and dry (e.g., dry-pit mining). Climate change may have direct and indirect effects that are analogous to anthropogenic activities, such as water withdrawals or perennialization. Inadequate public awareness and scientific understanding have hampered development of management techniques to effectively prevent, reduce, or mitigate the impacts of anthropogenic alterations on IRES ecosystems.

Increased demands for water, and consequent impacts on IRES, will be unavoidable in the future. Moreover, threats to ecosystem services (e.g., carbon storage or release, biodiversity) of IRES are poorly documented. Consequently, effective conservation of these systems will require information about additive or synergistic effects of the multiple threats that these unique systems face. Future efforts to better understand the socioeconomic value of ecosystem services may increase public awareness of the importance of IRES (Chapter 5.2). Likewise, legislative mandates may be a key component of both conservation efforts and the basis for effective management of IRES (Chapters 5.3–5.5).

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